

# Sourav Minhas

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## EDUCATION

### Carleton University, Department of Computer Science

Sep 2022 – Apr 2027

*Honours Bachelor of Computer Science: Artificial Intelligence & Machine Learning*

*Ottawa, Ontario*

- **Awards & Achievements:** Dean's Honour List, VRC Robotics Design & Build Award
- **Minor, Standing & GPA:** Mathematics, Fourth Year, 10.0/12.0 (A-)

## WORK EXPERIENCE

### AMD

Sep 2026 – Present

*AI Tooling Intern*

*Markham, Ontario*

- Scheduled to work across firmware and systems engineering, focusing on hardware-software **ML** design and integration

### RBC

May 2026 – Aug 2026

*Quality Engineer Intern*

*Toronto, Ontario*

- Engineered an internal AI knowledge assistant that retrieved **Confluence** data through **MCP**, processed context using **BM25** and semantic search, and synthesized responses with **GPT-4o** for various engineering and troubleshooting queries
- Extended into an **LLM-powered** test generation pipeline that combined **Jira** requirements with retrieved domain context to generate **Gherkin** test cases and publish them to **qTest**, adopting a more left-shift approach with broader test coverage
- Developed **Playwright** automation workflows enhanced with **agentic AI** to generate, maintain, and self-heal test cases across POS reporting and transaction-search systems, improving overall regression coverage and reducing manual tests

### Nokia

Jan 2026 – Apr 2026

*Automation Engineer Co-op*

*Ottawa, Ontario*

- Led the full software development lifecycle of a **Go**-based firewall feature for the **NEDR** platform, from technical design and implementation to documentation and reviews, enabling **IPv4/IPv6 CIDR** whitelisting, blacklisting, and alerting
- Improved unit testing workflows for containerized services using **Java**, **JUnit**, and **Spring Boot** by debugging failing test cases, validating system behaviour, and resolving security and automation-related issues across testing environments
- Supported **CI/CD** workflows using **Docker**, **Kubernetes**, **Jenkins**, **Helm**, and **Ansible**, supporting cloud and baremetal environments while collaborating within an Agile/Scrum development team through sprint planning and daily standups

*Software Support Co-op*

Sep 2025 – Dec 2025

- Resolved **Jira** tickets for the **NSP** platform involving network and device management, API calls via **Postman**, workflow issues, DR setups, database restores in **SQL/PostgreSQL**, and **IP** and **MDM** configurations across deployments
- Deployed **OpenStack** VMs hosting cloud-native applications, using **Kubernetes**, **Podman**, and **Linux** to troubleshoot systems, monitor services via **Grafana/Prometheus**, and validate endpoints, enhancing product reliability for customers
- Debugged and maintained platform code, implementing fixes in **Python** and **Bash**, documenting changes, and integrating updates into the NSP codebase via **GitLab** and **Jenkins**, improving system reliability and long-term maintainability

### Pepperdata

Jun 2025 – Aug 2025

*Software Engineer Intern*

*Toronto, Ontario*

- Developed scalable **PyTorch** workloads on **AWS** and **GCP** with **Kubernetes**, building reproducible training and benchmarking pipelines with **Docker** and **Jenkins** to support distributed ML experimentation and infrastructure automation
- Implemented internal tooling to benchmark **GPU** performance for machine learning workloads, running fine-tuning and inference experiments to collect utilization metrics, compare different configurations, and analyze multiple stress tests
- Built an **LLM-based QA** system that transcribed **200+** recordings with **Whisper**, processed the data through the **Gemini Enterprise Agent Platform**, and fine-tuned models using **GKE** and **EKS** to improve access to all internal knowledge

## TECHNICAL SKILLS

**Languages:** Python, C++, C, Java, Go, JavaScript, TypeScript, SQL, Bash, Scheme, Prolog

**Libraries:** PyTorch, TensorFlow, pandas, NumPy, scikit-learn, Matplotlib, Hugging Face Transformers, OpenCV

**DevOps & Cloud:** Docker, Podman, Kubernetes (OpenShift), Helm, Ansible, Jenkins, AWS, GCP, GitHub Actions

**Frameworks & Tools:** React, Node.js, Spring Boot, Playwright, JUnit, Git, GitHub, GitLab, Postman, SQLite, OpenStack